Fundamentals of Programming (CS-114)

Printing Data and Concatenation

Dr. Aamir Alaud Din

The pathway to Machine Learning and Artificial Intelligence

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Recap

- Python is a free programming language licensed under PSF which can be downloaded, distributed, installed, and used free (libre).
- Python packages are released by PyPI.
- Anaconda provides python with several packages incorporated which is very useful specially for the newbies.
- Python has different working environments including interactive mode, text and terminal, and IDE arrangements.
- The installer provided by Anaconda and Spyder IDE (because it has MATLAB like layout as well) are highly recommended for scientific computing.

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Objectives

After taking this lecture and studying, you should be able to

- Describe in your own words different data types in human communication.
- Describe different data types in python in general and numeric data types in particular in python, in your own words.
- Describe in your own words, the use of assignment operator.
- Describe in your own words, why variables are used in python programming.
- Describe the rules for naming variables in python in your own words.

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The Why Section

- In our daily life, we speak, read, write, and listen.
- In speaking, reading, writing, or listening, do we have data types?
- If yes, what data types we use in speaking, reading, writing, or listening?
- To what physical attributes, the data types are linked (reason of data types)?
- Does python programming (in fact all computer programming languages) also have the same data types?
- Are there additional data types in python?

The Why Section

- If yes, which additional data types python programming possesses and what are the reasons of having additional data types?
- Clinical digital thermometers sense the body temperature of a patient.
- They also display the temperature on the basis of which doctors take actions.
- It means displaying data in some cases is very important.
- When we perform computations, we use data and the outcome of computations is also data.
- The output data need to be displayed for authorities to take action on the basis of outcome of the computation.

The Why Section

Figure 1. Clinical digital thermometer with digital display.

The Why Section

- Does python programming has the ability to display data on the standard output display device (LCD display)?
- If yes, how to display data on standard output display device.
- Suppose you write a computer program of 100 lines and you perform the following computation on line number 8 of the program.

```
2.5*10 + 0.5*9.8*10**2
```

- How can you use the result of this computation on line number 97 of the program?

The Why Section

- Your python class instructor tells you that you can store this value in a variable and use the variable on line 97 of the program.
- How to store the output of the computation in a variable (assign output of computation to a variable)?
- Suppose you assign the output of the computation to the variable `computed out`.
- You use `computed out` on line number 97, execute the program and bugs pop up in front of you.
- The instructor tells you that your selection of variable name is wrong because you didn't follow the rules for naming variables.

The Why Section

- Are there some rules for naming variables in python?
- If yes, what are the rules for naming variables in python?
- What are python keywords?
- Why python keywords shouldn't be used as variables in python?
- We answer to all these questions shortly.

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Data Types

- Once I decided to bring the soft drink category ice cream soda, sold under the brand name of Pakola.
- I decided to purchase twelve (12) cans plus a two and a quarter of a liter (2.25 L) of the drink.

Figure 2. Pakola ice cream soda with 12 can (left) and a 2.25 L bottle (right).

Data Types

- I went to the store and said, “Give me 12 cans and a 2.25 L of Pakola ice cream soda.”
- In the above quote statement, 12 and 2.25 are the numbers and everything else is a text data, you already know.
- You also know that 12 is a whole number and 2.25 is a decimal point number.
- My question is, “To what attributes the whole numbers and decimal point numbers, in general, are linked?”
- Let’s start by breaking the flow of the lecture (in fact the lecture is in flow), if you count the number of cars in the car parking, can you say 20.25 cars or 20.93 cars?

Data Types

- Definitely, no.
- You will tell a whole number like 19 or 23 etc.
- Now, another question, “Is there any fractional error in the number of cars, you counted in the parking?”
- Definitely, no.
- It means, when we count objects, we use whole numbers and there is no fractional error in the whole numbers.
- So, just like real world, when we have to count in programming, we use int data type.

Data Types

Whole NUmbers or Integers

Whole numbers are attributed to counting of objects with no fractional error just like in counting the number of cars in the car parking or asking to give 12 cans of ice cream soda to the seller in the store. In computer programming, whole numbers are termed as integers represented by `int`.

- Suppose, you are performing an experiment in the lab and you measure the mass of sodium chloride.
- You say the mass of the salt is 2.5 grams.
- Can you claim that the salt is exactly (100%) 2.5 grams?

Data Types

- Definitely, no.
- Let's play with very simple computations of $\left(\frac{1}{50}\right) \times 50$ and $\left(\frac{1}{49}\right) \times 49$.
- We expect $\left(\frac{1}{50}\right) \times 50 = \left(\frac{1}{49}\right) \times 49 = 1.0$.
- But, computers don't give the output as expected.
- The answer lies in the output of
  1. $\frac{1}{49}$ and $\frac{1}{50}$
  2. Displaying the results to at least 19 places of decimal.

Data Types

- As we see, the fractional numbers are accurate up to 18 places of decimal in python and the 19th place has error.
- In the real physical world also, fractional numbers are not 100% certain.
- It means, when we measure things, we use fractional numbers which always have error.

Real Numbers or Floating Point Numbers

Fractional numbers are attributed to the measurements which always have error just like in the measurement of sodium chloride in the lab or the measurement of volume of ice cream soda filled in a bottle by the company. In computer programming, fractional data is termed as floating point numbers represented by `float`.

Data Types

- Any kind of English (or other) text is called the text data type in real world whereas in python programming it is termed as string data type represented by `str`.
- The `int` and `float` data types fall under the category of numbers on which arithmetic operations are possible whereas impossible on `str` data type.
- In the Figure 3 shown below, we can see that only three data sub-types are in use in human communication which are `int`, `float`, and `str`.
- Python has additional data types because these data types belong to science and engineering (all humans are not scientists and engineers) and to work with science, additional data types are required.

Data Type

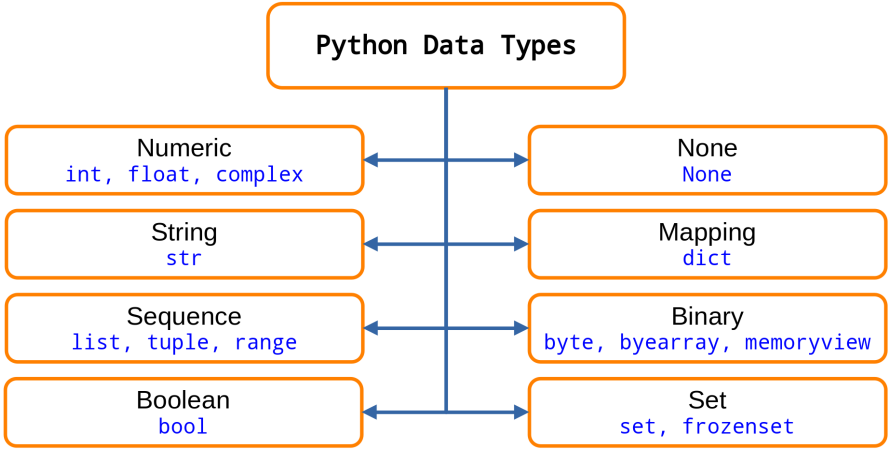

Figure 3. Python data types.

Data Types

- In the same way, human also have several data types which cannot be used in computers e.g., communication in symbolic form etc.
- In other words, data types are domain specific and therefore human communication, animal communication, and computers have different data types of their own domain.
- In the data types shown, we will touch numeric, string, sequence, boolean, and None data types as they are of our use and others will be rarely used.
- The data types which we will study cover the area of scientific computing related to data and models while other data types are specially for computer hardware etc.

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Numeric Data Types

- Numeric data types are of following three types.
  1. Integers
  2. Floating Point Numbers
  3. Complex
- Now, we cover these data types one by one.
- These data types are the most used data types in science and a solid understanding will help write good computer programs.
- They are not only important in science and engineering, but also in all other domains of work.

Numeric Data Types

1. Integers

- Integers are the whole numbers.
- They are attributed to counting of objects.
- Arithmetic operations are applicable and exact.
- Examples of integers are 29, -40, and 11 etc.
- Thousand, million, and billion separation are application with the use of underscore symbol e.g., 299,792,458 can be written as `299_792_458` in python.
- The python returns `int` data type for integers.

Numeric Data Types

- Below we show some examples on integers.

```
print(98)
print(type(98))
```

```
98
<class 'int'>
```

```
print(-15)
print(type(-15))
```

```
-15
<class 'int'>
```

Numeric Data Types

2. Floating Point Numbers

- Floating-point numbers are decimal point numbers which are attributed to measurements.
- Arithmetic operations are applicable and accurate to 18 places of decimal in python.
- Examples of floating point numbers are 3.14159, 8.314, and −2.00233184123 (electron g-factor) etc.
- They can also be represented in scientific notation e.g., $6.62607015 \times 10^{-34}$ and 6.67430×10^{-11} etc.
- The python returns `float` data type for floating point numbers.

Numeric Data Types

- Some examples of floating point numbers are shown below.

```
print(2.5)
print(type(2.5))
```

```
2.52
<class 'float'>
```

```
print(0.00001)
print(type(0.00001))
```

```
1e-05
<class 'float'>
```

Numeric Data Types

3. Complex

- Complex data type consists of two part namely real part and imaginary part.
- They are attributed to complex numbers.
- Arithmetic operations are applicable as per mathematics rules for complex numbers.
- Examples of complex numbers are $3 + 2i$ and $4i$ etc.
- In python, they are represented as `3+2j` and `4j` etc.
- The python returns `complex` data type for complex numbers.

Numeric Data Types

- An example of a complex data type is shown below.

```
print(3+2j)
print(type(3+2j))
```

```
3+2j
<class 'complex'>
```

- The real part of a complex number can be called by the following line of code in python.

```
print((3+2j).real)
```

```
3.0
```

Numeric Data Types

- The imaginary part of a complex number can be called by the following line of code in python.

```
print((3+2j).imag)
```

```
2.0
```

Caution!

The symbol `>>>` is known as python prompt and you don't need to type it. Always type the text after the python prompt symbol, for example, `3+2j` only instead of `>>> 3+2j`.

Numeric Data Types

- We know that complex numbers always exist in conjugate pairs.
- Python has the facility to generate the conjugate pair of the given complex number automatically.
- The conjugate pair can be called by the following line of code on python prompt.

```
print(3-1.2j).conjugate()
```

```
(3+1.2j)
```

Numeric Data Types

Methods and Attributes

In calling real and imaginary parts, we used the keywords `real` and `imag` (no parenthesis after keyword), however, we used `conjugate()` keyword (parenthesis after keyword). The keywords `real` and `imag` are called the attributes while the keyword `conjugate()` is called a method. We will learn about these two terms in the upcoming lectures.

- The keywords where some input data might be necessary, python uses parenthesis after those keywords which are known as functions or methods.
- The keyword where input data is impossible, python doesn't use parenthesis and these keywords are called the attributes.

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Strings

- Strings are any English alphabets, words, sentences, or any other text (including numbers) enclosed with single or double quotes.
- Python represents string data type as `str`.
- Mathematical operations are not possible on string data type.
- Mathematical operations '+' and '*' are valid but they have different output than mathematical outputs.
- The '+' operation joins strings and '*' replicates the string the number of times the number is multiplied.
- Examples below clarify these two operations.

Strings

```
print('Aamir ' + 'Alaud' + ' Din')
print(3*'Aamir')
```

```
'Aamir Alaud Din'
'AamirAamirAamir'
```

- Strings are an important part of computer programming.
- Dealing with strings is an individual topic.
- We will study this topic in the upcoming lectures.

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Data Type Conversion

- The functions `int()`, `float()`, `complex()`, and `str()` are used to convert a given data type into `int`, `float`, `complex`, and `str` data types, respectively.
- However, conversion depends on the input data type and the function used.
- Of all the data types discussed above, only `str()` is the only function which can convert any data type into `str` data type.
- It is because, it has to close the input data into single quotes only to convert the input data to `str` data type.
- Following tables clarify the possiblity of conversion of data types.

Data Type Conversion

Table 1. Conversion of one data type into other data types.

Input Data Type	Function Used	Output Data Type
`int`	`int()`	`int`
	`float()`	`float`
	`complex()`	`complex`
	`str()`	`str`
`float`	`int()`	`int`
	`float()`	`float`
	`complex()`	`complex`
	`str()`	`str`

Data Type Conversion

Table 1. Continued.

Input Data Type	Function Used	Output Data Type
`complex`	`int()`	Not possible
	`float()`	Not possible
	`complex()`	`complex`
	`str()`	`str`
`str`	`int()`	`int` (Only if string has digits only)
	`float()`	`float` (Only if string has digits only)
	`complex()`	`complex` (Only if string is of complex format)
	`str()`	`str`

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Assignment Operator

- Consider the mathematical equations

$$x = 5$$

$$x = x + 2$$

- The equation $x = x + 2$ is invalid because using $x = 5$ will not validate the equation $x = x + 2$.
- Now consider the following python statements.

```
x = 5
x = x + 2
```

- The above python statements are correct.

Assignment Operator

- The first line of code stores (assigns) the value 5 (right hand side) to x (left hand side).
- The second line computes the right hand side (x + 2) and stores (assigns) the result (7) to the left hand side (x).
- In python the symbol = is not the equality sign but, it is called the assignment operator.
- Its job is to assign the result of right hand side to the left hand side.
- So, in any python statement containing assignment operator, python computes the right hand side and assigns its result to the left hand side which is called a variable.

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Variables

- A variable is an alphanumeric combination which can be used in place of the data which has been assigned to it.
- However, these alphanumeric combination must be used in the light of some rules which will be discussed shortly.
- Any data (integer, floating point number, complex, or string) can be assigned to the variable.
- The string data must be assigned within single or double quotes like `x = 'Aamir'` or `x = "Aamir"`.
- An example of using variables is shown below.

Variables

```
pencil_length = 5.8 # inch
pencils = 5
total_length = pencils*pencil_length
print(total_length)
```

```
29.0
```

Rules for Naming Variables

- Although, variable names are a combination of alphanumeric characters, however, the following rules for naming variables must be followed.

Variables

1. The variable names are case sensitive e.g., bonds and Bonds are two different variable names.
2. Variable names can contain letters (26 uppercase and 26 lowercase), digits (0~9), and underscore (_) sign.
3. The variable name must not start with a digit.
4. The variable must not be a python keyword.
5. There must not be any space in the variable name.

Variables

- Python keywords can be found by entering the command `help('keywords')` at the python prompt.
- Keywords must not be used as variable names in any python program.
- Python may consider variable names as reserved keyword which python uses which may lead to wrong results.
- Always use meaningful names for variables which give an idea about the variable and the program.
- For example, if dealing with a molecular simulation, `bond_length` will be more meaningful than `x`.
- Rules for naming python files are the same as for variables.

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The Side Dish: Immutability and Identity

- Once a python variable is defined (assigned with some data), its value cannot be changed unless reassigned with some other value.

```
x = 5
y = x
```

- The above code assigns a value of -5 to x and y.
- Next, we reassign some value to x as show below.

```
x = 13
```

- Will this reassignment change the value of y also?

The Side Dish: Immutability and Identity

- The value of y will not chage.
- Before reassignment of x, the location addresses of x and y are the same.
- After reassignment of x, the location addresses of x and y become different.
- See the program on the location addresses of both variable before and after reassignment of x.
- It means, python objects are immutable.
- Python objects don't change once created, however, variable names may be reassigned to refer to a different object.

The Side Dish: Immutability and Identity

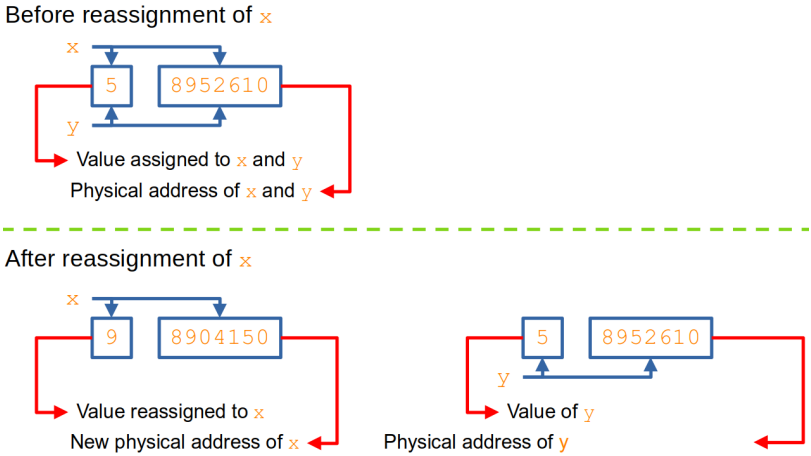

Figure 4. Reassignment of values doesn't change the value of first variable which is immutability.

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Summary

- Like data used among human beings, python programming also uses the similar data in programming.
- Python also has additional data types related to the field of computers.
- The numeric data types in python are `int`, `float`, and `complex`.
- Strings are the text data types.
- Some of the data types can be converted into other data types.
- Assignment operator assigns right hand side data values to the variables on left hand side in a python statement.
- Naming variables must be as per the rules.

Thanks!